

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

The Quantum World

KKKC6353 Nanoelectronic Devices — Lecture 2, Document 1 of 3

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How to use this document

This document is a gentle first meeting with quantum mechanics. You do not need to have studied it before. The aim is that you come away with a clear *picture* of a handful of ideas, because everything else in Lecture 2 is built on them:

- tiny things behave as **waves** as well as particles;
- a wave that is **trapped** can only have certain energies;
- the **smaller** the trap, the more widely spaced those energies are.

Document 2 uses these ideas to explain quantum wells, wires and dots and the devices built from them. Document 3 describes how they are made.

Each section opens with a blue **Key ideas** box and closes with a few **check questions**. Green **Going further** boxes hold the mathematics behind the pictures. They are optional: read them if you want to see where the numbers come from, skip them if you do not. Boxes marked **A common misconception** point out ideas that sound right but are not.

1 Small enough to change colour

> **Key ideas**
>
> A quantum dot is a crystal of semiconductor only a few nanometres across.
>
> Dots made of the same material glow different colours depending only on their size.
>
> Ordinary physics cannot explain this. Quantum mechanics can, and this document shows how.

Shine ultraviolet light on the row of vials below and each one glows a different colour, from violet to red. Every vial holds the same material. The only difference is the size of the tiny crystals inside: the smallest glow violet and the largest glow red.



Figure 1.1. *Quantum dots of one material, different sizes, glowing under ultraviolet light. The smallest dots glow violet and the largest red. Photograph: Antipoff, Wikimedia Commons (CC BY-SA 3.0).*

That is strange. A large lump of a material has one colour; cutting it in half does not change it. Yet when the crystals are only a few nanometres across, about ten to fifty atoms wide, size suddenly matters. These crystals are called **quantum dots**, and they are already in television screens and in medical imaging.

To see why size controls colour, we need some new ideas about how very small things behave. By the end of Section 6 the colours of these vials will make sense.

Watch. A two-minute introduction from the US National Institute of Biomedical Imaging and Bioengineering (NIBIB): *What are quantum dots?*, youtube.com/watch?v=LlPDyl53rZA.

2 Two worlds: classical and quantum

> **Key ideas**
> Classical physics describes the everyday world very well.
> It fails for things as small as atoms and electrons.
> The atom itself is the clearest example: classical physics says it should collapse.

2.1 The physics of the everyday

The physics you learned first, often called **classical physics**, describes the world at human scale: falling apples, moving cars, rocket engines. It rests on three pillars:

Pillar	What it describes
Newtonian mechanics	motion and forces
Electromagnetism	electricity, magnetism and light
Thermodynamics	heat, energy and engines

Classical physics pictures the universe as a predictable clockwork machine. If you know where everything is and how it is moving now, you can work out what happens next. For everyday objects this works superbly.

It breaks down at the extremes. For things moving close to the speed of light we need relativity. For things as small as atoms and electrons we need **quantum mechanics**, and the rules there feel very different. The physicist Richard Feynman, who won the Nobel Prize for his work on it, once said: "I think I can safely say that nobody understands quantum mechanics." Do not worry if parts of it feel odd. It feels odd to everyone; what matters is that it predicts what experiments find, every time.

2.2 Why the planetary atom fails

A popular picture of the atom is a tiny solar system: electrons orbiting a nucleus like planets around the Sun. Classical physics says this cannot last. An electron moving in a circle is constantly changing direction, and a charge that changes direction gives off light. Giving off light costs energy, so the electron should spiral inwards and crash into the nucleus in about a hundred-billionth of a second.

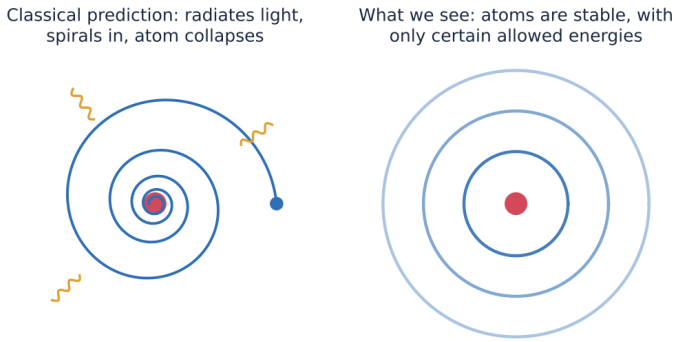


Figure 1.2. *Left: what classical physics predicts for an orbiting electron. It radiates light, loses energy and spirals into the nucleus. Right: what we actually see. Atoms are stable and their electrons have only certain allowed energies. Figure made for this course.*

Atoms do not collapse. You are made of them, and they have lasted billions of years. So at the scale of an atom, classical physics is simply wrong. We need a new set of rules.

Check your understanding

1. Name the three pillars of classical physics.
2. In your own words, why does classical physics predict that atoms should collapse?

3 Light: a wave or a stream of particles?

> **Key ideas**
> Light sometimes behaves as a stream of particles, called photons.
> Light sometimes behaves as a wave.
> Both are true. Which one you see depends on the experiment.

Classical physics insists that a thing is either a **wave**, spread out and carrying energy smoothly, or a **particle**, a small lump at one place. Two experiments with light show that nature does not respect this split.

3.1 Evidence that light is made of particles

Shine light on a clean metal surface and, under the right conditions, electrons are knocked out. This is the **photoelectric effect**. The surprise is *which* light does it.

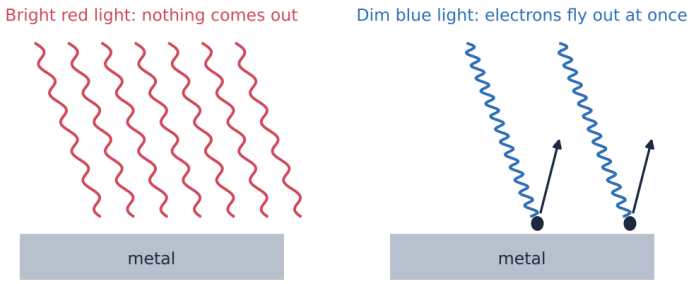


Figure 1.3. *The photoelectric effect. Left: red light, however bright, knocks out no electrons from this metal. Right: even dim blue light knocks electrons out at once. Figure made for this course.*

If light were a smooth wave, a brighter beam would carry more energy and should eventually shake electrons loose whatever its colour. That is not what happens. Below a certain colour nothing comes out, however bright the light. Above it, electrons fly out instantly, even when the light is very dim.

Einstein explained it in 1905. Light arrives as separate packets of energy, called **photons**, like a stream of tiny bullets. Each photon gives its energy to one electron. A blue photon carries more energy than a red one, enough to free an electron; a red photon does not, and sending more red photons does not help. A brighter beam just means more photons, not stronger ones.

3.2 Evidence that light is a wave

Now send light through a very narrow gap. If light were a stream of tiny balls it would go straight through and make one sharp spot on the screen behind. Instead it spreads out on

the far side, as ripples on a pond do when they pass through a gap in a wall. This spreading is called **diffraction**, and it is a sure sign of a wave.

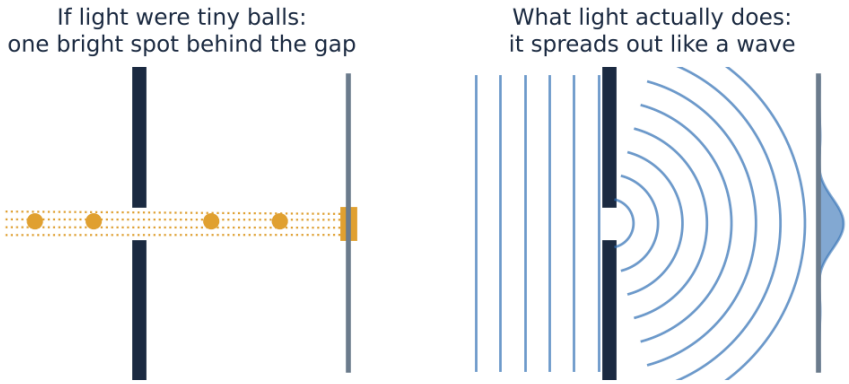


Figure 1.4. *Left: if light were tiny balls, it would make one spot behind the gap. Right: light actually spreads out after the gap, as a wave does. Figure made for this course.*

So light is both. It travels like a wave and arrives like a particle. This is called **wave–particle duality**.

> **Going further: the energy of a photon (optional)**
>
> A photon of frequency f carries energy $E = hf$, where $h = 6.63 \times 10^{-34}$ J s is Planck's constant. Higher frequency means bluer light and more energy per photon.
>
> It is often handier to use the wavelength λ. In convenient units, E (eV) $\approx 1240/\lambda$ (nm). Red light at 650 nm carries about 1.9 eV per photon; blue light at 450 nm carries about 2.8 eV.
>
> An electron leaves the metal only if one photon brings more energy than the metal's *work function*, the energy holding the electron in. For caesium that is about 2.1 eV, so blue light frees electrons from it and red light does not.

Check your understanding

1. Why does making red light brighter not knock electrons out of the metal?
2. What does diffraction tell us about light?

4 Matter waves

> **Key ideas**
>
> Electrons, and in fact all moving things, also behave as waves.
>
> The heavier and faster something is, the shorter its wavelength.
>
> An electron's wavelength is a few nanometres, the size of a modern transistor, so its wave nature matters in devices.

4.1 De Broglie's idea

If light, which we thought was a wave, can behave as particles, could particles behave as waves? In 1924 Louis de Broglie suggested exactly that. Every moving object has a wavelength, and it is set by its **momentum**, its mass times its speed:

$$\lambda = \frac{h}{\text{momentum}}$$

The number h on top is tiny, so the wavelength is tiny unless the momentum is tiny too. The rule is simple: **the bigger the momentum, the shorter the wavelength**.

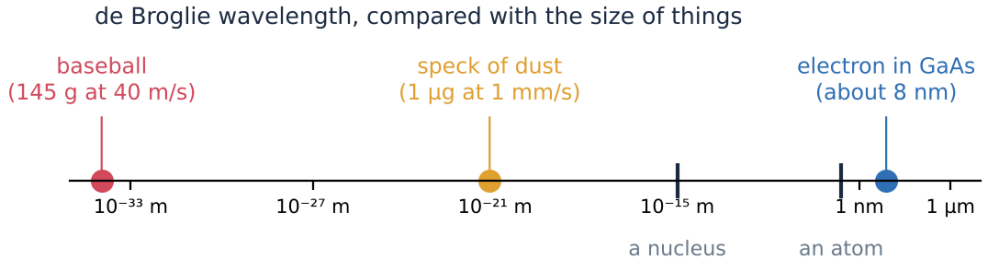


Figure 1.5. *The de Broglie wavelengths of a thrown baseball, a drifting speck of dust and an electron in a semiconductor, on a scale where each step is a factor of a million. Figure made for this course.*

A thrown baseball has a wavelength far smaller than a nucleus, so no experiment could ever see its wave nature. That is why you and I do not diffract through doorways. An electron is so light that its wavelength is a few nanometres, about the size of the features in a modern chip. That is why the wave nature of electrons matters in nanoelectronics, and why it shows up first in the smallest devices.

4.2 Electrons interfere one at a time

The most striking evidence comes from sending electrons, one at a time, towards two narrow slits. Each electron arrives at the screen as a single dot, like a particle. But as thousands arrive, the dots build up into stripes, the interference pattern of a wave passing through both slits at once.

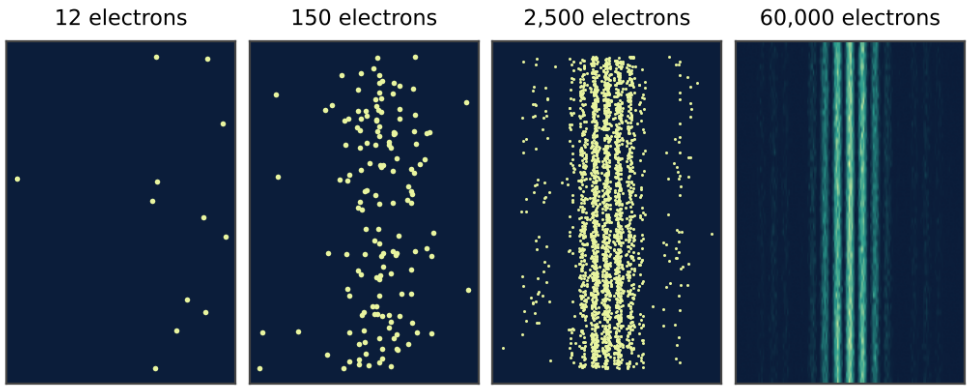


Figure 1.6. *Electron interference built up one electron at a time, simulated after the 1989 experiment of Tonomura and colleagues. Each electron lands as a dot, but where the dots fall follows the stripes of a wave. Figure made for this course.*

The wave tells us *where an electron is likely to land*. The particle is what we actually detect. This idea, a wave that sets the chances and a particle that turns up, is the heart of quantum mechanics.

> **Going further: the numbers (optional)**
> De Broglie's relation is $\lambda = h/p$, with momentum $p = mv$.
> Baseball: $m = 0.145$ kg, $v = 40$ m/s, so $\lambda = 6.63 \times 10^{-34}/(0.145 \times 40) \approx 1 \times 10^{-34}$ m.

> Electron in GaAs at room temperature: it moves as if its mass were $0.067\,m_0$ (Section 7) with a typical thermal energy of about 0.04 eV, which gives $\lambda \approx 8$ nm.

Check your understanding

1. Why can we never see the wave nature of a baseball?
2. In the two-slit experiment, what part of the result looks like a particle and what part looks like a wave?

5 The Schrödinger equation

> **Key ideas**
>
> The Schrödinger equation is the rule that tells us what shape an electron's wave takes.
>
> At heart it is just energy bookkeeping: kinetic energy plus potential energy equals total energy.
>
> The wave itself is not seen. Its square tells us where the electron is likely to be found.

5.1 Energy bookkeeping

If an electron is a wave, we need a rule for what that wave looks like in a given situation. In 1926 Erwin Schrödinger wrote one down. It looks fearsome, but it says something you already know:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$$

kinetic energy (how much the wave wiggles) | potential energy (the landscape) | total energy

Figure 1.7. *The Schrödinger equation read as an energy balance. Figure made for this course.*

kinetic energy + potential energy = total energy

The wave is written ψ (the Greek letter psi). The first term is the kinetic energy, the energy of motion, and it is larger when the wave wiggles more tightly. The second term is the potential energy, set by the landscape the electron sits in: walls, hills and valleys made by the materials around it. The right-hand side is the total energy.

Solving the equation means finding the wave shapes that balance this bookkeeping at every point. For most landscapes only certain shapes manage it, and each one comes with its own energy. Those are the **allowed energies**, and finding them is what the Schrödinger equation is used for: in atoms, in molecules and in the nanostructures of this lecture.

Watch. Up and Atom, *What is the Schrödinger equation, exactly?*, youtube.com/watch?v=QeUMFo8sODk. A friendly visual explanation of where the equation comes from.

5.2 What the wave means

The wave ψ is not something we can see or measure directly. What we can measure is where the electron turns up. The rule, due to Max Born, is that the chance of finding the

electron at a point follows $|\psi|^2$, the size of the wave squared. Where the wave is large, the electron is often found; where the wave is zero, it is never found.

5.3 A built-in fuzziness

Waves bring one more surprise. A wave that is squeezed into a small region has to be made of many different wavelengths mixed together, and a wave with one clean wavelength has to be spread out. Since wavelength is tied to momentum, this means we cannot know exactly both where an electron is and how fast it is moving. This is **Heisenberg's uncertainty principle**. It is not a fault of our instruments: it is how waves are. We will meet it again in the next section, where it stops a trapped electron from ever sitting still.

> **Going further: the equation itself (optional)**
>
> For an electron of mass m moving along one direction x in a potential energy $V(x)$, the time-independent Schrödinger equation is $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\,\psi = E\,\psi$, with $\hbar = h/2\pi$.
>
> The second derivative $d^2\psi/dx^2$ measures how sharply the wave curves. A tightly wiggled wave curves sharply, which is why it carries more kinetic energy.
>
> Born's rule: the probability of finding the electron between x and $x + dx$ is $|\psi(x)|^2\,dx$, and the total probability over all space is 1.
>
> Heisenberg's principle: $\Delta x\,\Delta p \geq \hbar/2$, where Δx and Δp are the spreads in position and momentum.

Check your understanding

1. Write the Schrödinger equation in words, as an energy balance.
2. What does $|\psi|^2$ tell us?

6 An electron in a box

> **Key ideas**
>
> A trapped electron can only have certain energies, just as a guitar string can only play certain notes.
>
> The smaller the box, the further apart those energies are.
>
> A trapped electron can never be completely still.

6.1 Only certain notes fit

Pluck a guitar string and it plays a note. The string is pinned at both ends, so it can only vibrate in shapes that are still at the ends: one hump, two humps, three humps, and so on. Shapes in between would need the ends to move, and they cannot. That is why a string plays a definite note and its overtones, not any pitch you like.

An electron trapped in a tiny box behaves the same way. Its wave has to fit inside the box and fall to zero at the walls, so only a whole number of half-waves will do. We label these shapes with a whole number n = 1, 2, 3, … Each shape has its own energy, and no energies in between are allowed. This is what *quantised* means.

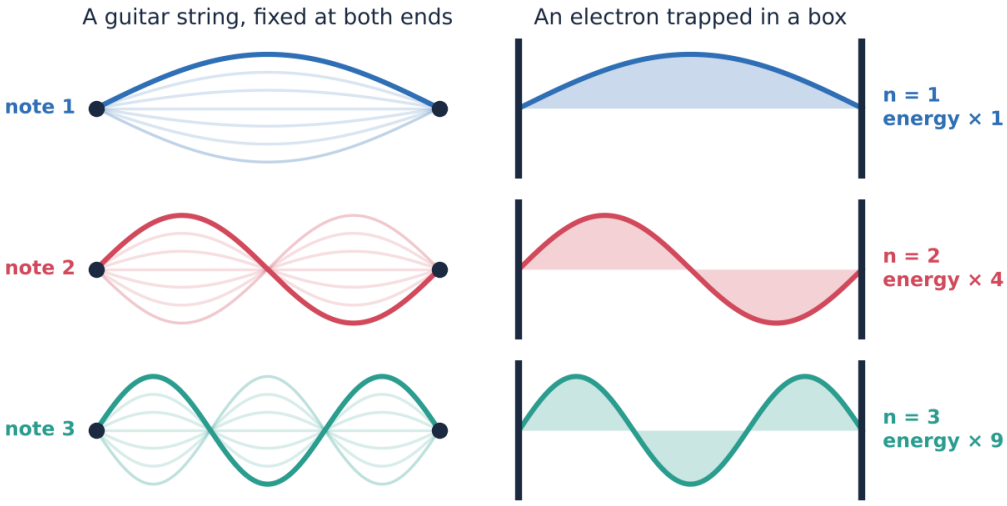


Figure 1.8. *Left: a guitar string pinned at both ends can only vibrate with one, two, three … humps. Right: an electron in a box is the same story. Only whole numbers of half-waves fit, and each one has its own energy. Figure made for this course.*

The wave with more humps is more tightly wiggled, and a more tightly wiggled wave carries more energy. Two humps cost four times the energy of one; three humps cost nine times. The energies climb as 1,4,9,16, …

6.2 What sets the energies

Two things decide how far apart the allowed energies are.

The size of the box. Squeeze the box and every wave has to wiggle more tightly to fit. Halve the width and every energy goes up four times. This is the most important fact in the whole lecture: *in a nanostructure, size sets energy*.

How heavy the particle is. A lighter particle has more widely spaced levels. Inside a semiconductor an electron moves as if it were lighter than a free electron (about one-fifteenth of the mass in GaAs, Section 7), so semiconductor boxes show these effects at sizes of a few nanometres to a few tens of nanometres.

6.3 It can never sit still

Notice that there is no $n = 0$. A wave with zero humps is no wave at all: it would mean there is no electron in the box. So the lowest energy a trapped electron can have is the $n = 1$ energy, and it is not zero. The everyday intuition, that a particle in a box could simply rest on the floor, fails.

There is a second way to see it. If the electron were perfectly still, we would know its speed exactly (zero), and we would also know it is somewhere inside the small box. The uncertainty principle of Section 5 does not allow both to be known that well at once. Confining a particle forces it to keep moving.

6.4 Where is the electron?

The wave itself, written ψ ("psi"), is not something we can see. What we can measure is where the electron turns up when we look. The chance of finding it at each point follows $|\psi|^2$, the wave squared.

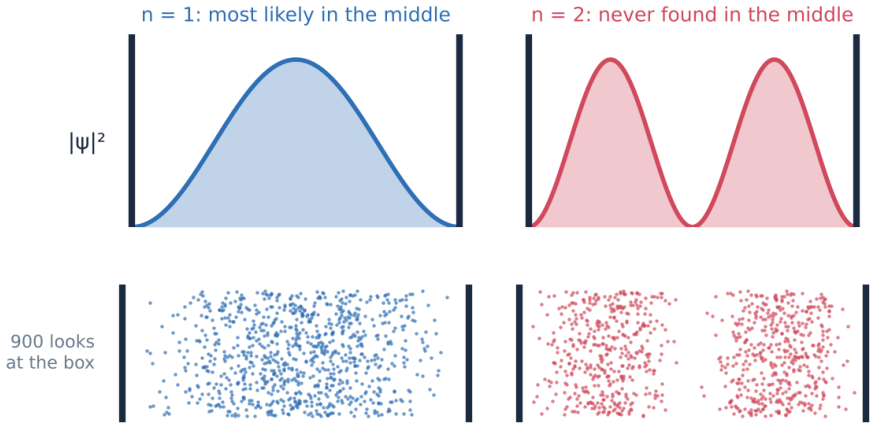


Figure 1.9. *Top: where the electron is likely to be, for the two lowest states. Bottom: 900 imaginary measurements, each one a dot where the electron was found. In the n = 2 state it is never found in the middle of the box. Figure made for this course.*

In the lowest state the electron is most often found near the middle. In the second state it is never found exactly in the middle, even though it is found on both sides. A classical ball rolling back and forth would turn up everywhere equally. This is a purely wave-like result.

> **A common misconception**
>
> "The electron is spread out like a cloud across the box." Not quite. Each time we look, we find the whole electron at one place. The curve tells us how the places are spread over many looks, as the dots in the figure show.

6.5 Real walls leak a little

A real box, such as a thin layer of one semiconductor inside another, has walls of finite height. The wave does not stop dead at such a wall: it leaks a short way into it and dies away. Because the wave has more room, the levels sit lower than in the ideal box (for a 10 nm GaAs box, 32 meV instead of 56), and a box only so deep holds only a few levels. The leaking wave is also the start of *tunnelling*, the subject of Lecture 3.

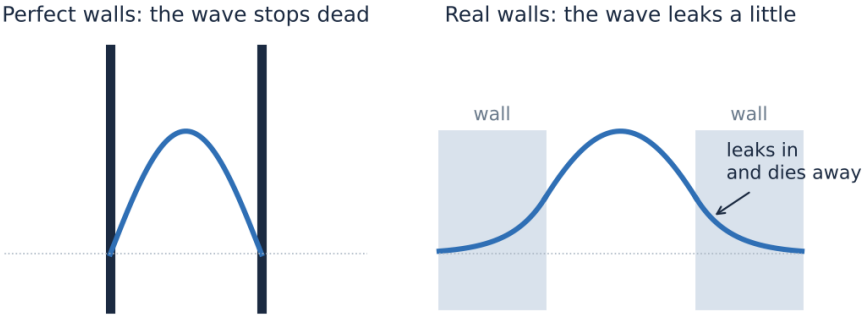


Figure 1.10. *With walls of finite height the wave leaks a short way into the walls. The levels sit lower than for perfect walls. Figure made for this course.*

6.6 Why this matters: colour from size

Put an electron in a box and give it energy, and it can drop from a higher level to a lower one by giving out light. The colour of that light is set by the gap between the levels, and the gap is set by the size of the box. A large quantum dot glows red; a small one, made of the same

material, glows blue. This is the quantum-dot colour we opened the document with, and it is now explained.

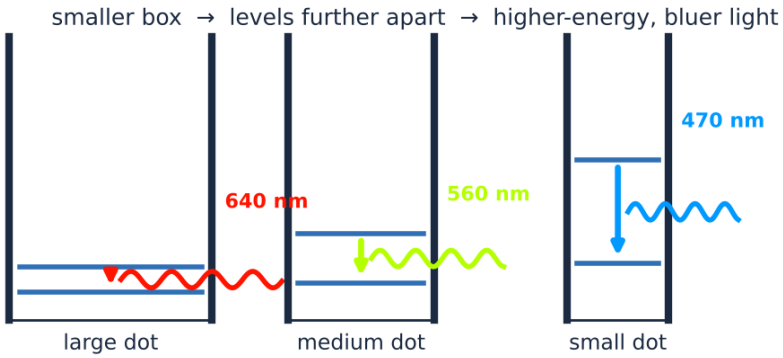


Figure 1.11. *Smaller box, levels further apart, higher-energy and bluer light. The same material gives different colours purely through size. Figure made for this course; the levels are drawn schematically.*

> **Going further: the numbers (optional)**
>
> For an electron of effective mass m^* in a box of width L with perfect walls, the allowed energies are $E_n = \frac{h^2 n^2}{8 m^* L^2}$, with $n = 1,2,3, \ldots$
>
> The n^2 gives the 1, 4, 9 pattern; the L^2 underneath gives "halve the box, four times the energy"; and m^* underneath gives "lighter particle, wider spacing".
>
> The wave shapes are $\psi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$, the same shapes as the guitar string.
>
> Example: an electron in GaAs ($m^* = 0.067\, m_0$) in a 10 nm box has $E_1 \approx 56$ meV with perfect walls. With the real walls of a GaAs/AlGaAs well (about 0.3 eV high) it drops to about 32 meV, because the wave leaks into the walls.

Check your understanding

1. A guitar string and an electron in a box both have only certain allowed "notes". What do the two have in common that causes this?
2. A box is made three times narrower. What happens to the lowest energy?
3. Give one reason, in your own words, why a trapped electron cannot have zero energy.
4. Two quantum dots are made of the same material. One glows green and the other red. Which is smaller, and why?

Explore. The animation *Particle in a box* (faizverse.github.io/nanoelectronic-devices/L2) lets you change the size of the box and watch the levels and waves respond.

7 From a box to a real semiconductor

> **Key ideas**
>
> Inside a crystal an electron moves as if it were lighter than a free electron.
>
> A box is made by sandwiching one semiconductor between others that act as walls.
>
> Quantum effects show up when the energy levels are further apart than the thermal jiggling at room temperature.

7.1 A lighter electron

Inside a semiconductor an electron is surrounded by a regular lattice of atoms. You might expect them to get in its way. Remarkably, a perfectly regular lattice does not scatter the electron at all. Its whole effect can be summed up by pretending the electron has a different mass, called the **effective mass**. In GaAs the electron behaves as if it were about one-fifteenth as heavy as a free electron; in silicon, about a quarter.

A lighter particle has a longer wavelength and more widely spaced levels in a box, so quantum effects appear in GaAs at larger sizes than in silicon. That is one reason so much of the physics in this lecture was first seen in GaAs and its relatives.

7.2 Making the box

In a real device, the box is made from materials. A thin layer of one semiconductor, such as GaAs, sandwiched between layers of another with a higher energy for electrons, such as AlGaAs, traps electrons in the thin layer. The outer layers are the walls; their height is set by the choice of materials, and the width of the box is the thickness of the layer. This sandwich is a **quantum well**, the first of the structures in Document 2.

7.3 When do the levels show?

Atoms and electrons are never perfectly still. At room temperature every electron is jostled with a typical energy of about 26 thousandths of an electronvolt (26 meV). Think of it as a blur on every energy level.

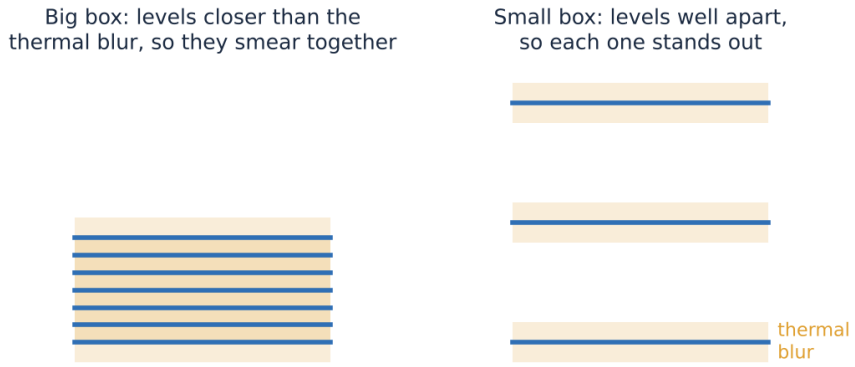


Figure 1.12. *Left: in a large box the levels are closer together than the thermal blur, so they merge and behave like a smooth range of energies. Right: in a small box the levels are well apart and each stands out. Figure made for this course; drawn schematically.*

If the levels in the box are closer together than this blur, they merge and the box behaves classically. If they are well apart, each level stands out and quantum effects are plain to see.

For electrons in GaAs, levels pull clear of the blur once the box is narrower than about 15 to 20 nm. That sets the size of the structures in the rest of this lecture: a few nanometres to a few tens of nanometres.

> **Going further: the numbers (optional)**
>
> The thermal energy at temperature T is $k_B T$, with k_B Boltzmann's constant. At 300 K, $k_B T \approx 25.9$ meV.
>
> For a GaAs well of width 10 nm with perfect walls, $E_1 \approx 56$ meV and $E_2 = 4E_1 \approx 225$ meV. The gap of about 170 meV is more than six times $k_B T$, so the levels are clearly separate.
>
> At 30 nm the same gap shrinks by a factor of nine, to about 19 meV, which is less than $k_B T$. The levels then blur together at room temperature, though they reappear when the sample is cooled.

Check your understanding

1. What does the effective mass stand in for?
2. Why do quantum effects appear in GaAs at larger sizes than in silicon?
3. A box has energy levels 5 meV apart. Would you expect to see quantum effects at room temperature? Why or why not?

8 Summary, and what comes next

1. **Very small things behave as waves as well as particles.** The photoelectric effect shows light arriving as particles; diffraction and electron interference show waves.
2. **The heavier and faster something is, the shorter its wavelength.** An electron's wavelength is a few nanometres, the size of today's devices.
3. **The Schrödinger equation is energy bookkeeping.** Its solutions are the allowed wave shapes and energies, and $|\psi|^2$ tells us where the electron is likely to be.
4. **A trapped wave has only certain energies**, like the notes of a guitar string, and it can never have zero energy.
5. **The smaller the box, the further apart the energies.** That is why quantum dots of different sizes glow different colours.
6. **Quantum effects show** when the levels are further apart than the thermal blur, which in practice means structures from a few to a few tens of nanometres.

Where Document 2 begins. A quantum well traps electrons in one direction and leaves them free in the other two. Trap them in two directions and you have a quantum wire; in all three, a quantum dot. Document 2 shows how each step changes what the electron can do, and how lasers, transistors, displays and nanotubes make use of it.

Words to know

Word	Plain meaning
Photon	A packet of light energy
Wave–particle duality	Light and matter behave as waves in some experiments and particles in others
de Broglie wavelength	The wavelength of a moving particle; shorter for larger momentum

Word	Plain meaning
Wavefunction, ψ	The wave that describes an electron; its square gives where it is likely to be
Quantised	Allowed only certain values
Ground state	The lowest allowed energy
Effective mass	The mass an electron appears to have inside a crystal
Quantum well, wire, dot	A structure that traps electrons in one, two or three directions

Sources and further reading

Gentle introductions

- R. P. Feynman, *QED: The Strange Theory of Light and Matter*, Princeton University Press, 1985. A famous non-mathematical account of light and electrons.
- R. P. Feynman, *The Character of Physical Law*, MIT Press, 1965. Chapter 6, "Probability and uncertainty", is the source of the quotation in Section 2.

Textbooks, for the Going further boxes

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Figures. Except for the photograph in Section 1, all figures were made for KKKC6353 (scripts in `figures-source/`).